

EVIDENCE.md — row-to-artifact map for the C4 numbers audit

Every R### row in RESULTS.md, and where to verify it. C4's brief is to trace each claim to a RESULTS row **or raw output**; this file is the index that makes that a lookup rather than a search.

Written because several rows previously cited `scratchpad/...` paths that existed only in a session-local temp directory and would have audited as UNSUPPORTED. Those artifacts are now committed under `evidence/`.

How to verify, in order of preference

- 1. **Re-execute** — the strongest check. `shadow_hadamard_challenge_PARTICIPANT.ipynb` is committed *with its outputs*, and re-running it end to end reproduces every Part A/B number (56/56 checkpoints, seed 2026, deterministic).
- 2. **Read the committed output** — `run_summary.json`, the notebook's own output cells, `data/*.csv`, `evidence/*.json`.
- 3. **Read the executed notebook of the side study** — `evidence/executed/*.ipynb`.
- 4. **Re-run the producing script** — `evidence/scripts/*.py`.

Row map

row	claim	primary evidence	re-runnable by
R001	Gate 0 PASS	— (run it)	<code>python scripts/gate0_benchmark</code>
R002	Checkpoint 2, 8 asserts	notebook cell 16 output	notebook re-run
R003	Checkpoint 3a, Challenge 1	notebook cell 24 output	notebook re-run
R004	Checkpoint 3b, sign conventions	notebook cell 26 output	notebook re-run
R005	Checkpoint 3c, measurement order	notebook cells 29/31 output	notebook re-run
R006	Checkpoint 4, Challenge 3	notebook cells 34/36 output	notebook re-run
R007	Checkpoint 5, Challenge 4	notebook cells 41/43 output	notebook re-run
R008	hardware job, survival 0.179 + withdrawn <Q> line (B04)	<code>hardware_jobs.json</code> ; <code>evidence/executed/executed_hw_analysis.ipynb</code>	<code>python hardware_run.py --fetch d9u95vs98n5s7392iao0</code> (live, reproducible)
R009	full sweep, 3456 circuits / 256,000 shots	<code>run_summary.json</code> ; notebook cell 48	notebook re-run
R010	Checkpoint 6, 8/8	<code>run_summary.json</code> ; notebook cells 51-59	notebook re-run
R011	Challenge 7 DFT baseline	notebook cell 65 output	notebook re-run
R012	Challenge 8 matrix pencil	notebook cell 70 output	notebook re-run
R013	Checkpoint 7, 8/8 + bootstrap sds	notebook cells 73-80 output	notebook re-run
R014	Challenge 11 noise, damping 0.342	notebook cells 91/93 output	notebook re-run
R016	Challenge 10 Krylov, E0 -1.5349	notebook cells 85/87 output	notebook re-run
R017	EXPLORATORY/CONFOUNDED label survival	<code>evidence/executed/executed_robustness2.ipynb</code>	<code>evidence/scripts/robustness_iter</code>
R018	hardware 2x2 COMPLETE (0.179/0.368/0.822/0.878)	<code>hardware_jobs.json</code> ; <code>evidence/hw_2x2_jobs.json</code>	<code>python hardware_run.py --fetch</code>
R019	12-seed study	<code>data/multiseed_summary.json</code> ; <code>data/*.csv</code> (24 files); <code>evidence/executed/executed_seedsweep.ipynb</code>	<code>evidence/scripts/seed_sweep.py</code>
R020	bootstrap validation ratios	same as R019	recompute from R013 + R019 sds
R021	ablation A: equivalence, 2x2 grid, cost	<code>evidence/ablation_results.json</code> ; <code>evidence/executed/executed_ablation_ab.ipynb</code>	<code>evidence/scripts/ablation_a_b.py</code>
R022	ablation B: resolution failure boundary	same as R021	same
R023	2-site hardware RETURNED : chi survival 0.962, genuine (Q) 0.957	<code>hardware_jobs.json</code> ; ideal-sim validation in <code>evidence/scripts/validate_2site.py</code>	<code>python hardware_run.py --fetch d9uk99k98n5s7392vhsg</code>
R024	<code>packing.py</code> unusable, third-party	—	<code>python -c "import packing"</code> (fails)
R025	transpile cost n=3 vs n=4, corrected	<code>evidence/scripts/transpile_cost_n3_n4.py</code> output (cross-checks itself against R018's 101)	<code>python evidence/scripts/transpile</code>
R026	4-site hardware RETURNED : survival 0.275 (vs teammate 0.587, same layout)	<code>hardware_jobs.json</code> ; ideal-sim validation in <code>evidence/scripts/validate_4site.py</code>	<code>python hardware_run.py --fetch d9ultm0u5hac73ahd9kg</code>
R027	rank selection (honest negative result)	<code>evidence/going_further_trio.json</code> ; <code>evidence/executed/executed_going_further.ipynb</code>	<code>evidence/scripts/going_further.py</code>
R028	measured chi/chi_Q correlation + correlated bootstrap	same as R027	same

row	claim	primary evidence	re-runnable by
R029	variance budgeting, 3 ^w rule vs measured	same as R027	same
R030	QPE ablation, 4/4 lines, no labels at any cost	evidence/qpe_ablation_result.json	python evidence/scripts/qpe_abl
R031	teammate's 6 real-hardware jobs (incl. ibm_miami), survival + phase-error finding	real_machine/*.json (raw job dumps, credential-checked); evidence/real_machine_analysis.json	python evidence/scripts/load_real_mach
R032	Track C eigenstate detective: calibration curve + detection power vs shots	evidence/track_c_result.json; evidence/track_c_console_output.txt	python evidence/scripts/track_c_eigenstate (~7 min)
R033	Track E consistency witness (3-leg validation) + product-state fidelity	evidence/track_e_result.json; evidence/track_e_console_output.txt	python evidence/scripts/track_e_consistency (~4 min)
R034	Track B (partial) A/B tester: identity 9e-15, overlap curves, gate cost	evidence/track_b_result.json; evidence/track_b_console_output.txt	python evidence/scripts/track_b_identity (~min)
R035	AQC-Tensor compression certified by the Track-B tester (nuanced: wins uncontrolled, loses controlled at n=3)	evidence/aqc_compression_result.json; evidence/aqc_console_output.txt	python evidence/scripts/aqc_compression (~min; needs pip install qiskit-a-tensor[quimb-jax])
R036	symmetry error detection: 0.000% ideal vs 1.650% real hardware	evidence/symmetry_detection_result.json; evidence/symmetry_detection_console.txt	python evidence/scripts/symmetry_error (~min)
R037	SKQD pipeline on hardware shots; config recovery removes 1 spurious eigenvalue	evidence/skqd_result.json; evidence/skqd_console_output.txt	python evidence/scripts/skqd_pipeline (~min)
R041	3-arm observable reconstruction; shadow-exclusive = non-commuting observables	evidence/observable_reconstruction.json; evidence/observable_reconstruction_console.txt	python evidence/scripts/observable_reconstruction (~3 min)
R040	protocol ladder; shadow inversion beats naive post-selection of same data 3.36x	evidence/protocol_escalation.json	python evidence/scripts/protocol_ladder (~3 min)
R039	direct-Z vs shadow: direct Z is 1.56x cheaper for Q, but blind to 4/9 terms of H	evidence/direct_z_ablation.json	python evidence/scripts/direct_z_ablation (~min)
R038	PROTOTYPE, next-phase — DOS from the same pipeline; weights become degeneracies	evidence/dos_prototype_result.json; evidence/dos_console_output.txt; evidence/dos_signals.npz	python evidence/scripts/dos_prototype (~min)

Derived quantities C4 should independently recompute

C4's brief (b) requires re-deriving ratios and CI widths from raw numbers. The ones that matter:

- **R020 bootstrap ratios** — across-seed sd (R019) ÷ bootstrap sd (R013), per line. Both sets of sds are printed in their rows.
- **R018 depth factor** — grid now COMPLETE. Depth (exact/trotter): marrakesh 0.822/0.179 = 4.59x, kingston 0.878/0.368 = 2.39x. Device (kingston/marrakesh): trotter 2.06x, exact 1.07x. Depth dominates on both devices; recompute all four from the table in R018.
- **R021 variance cost factor** — $0.1012 \div 0.0949 = 1.07x$. The denominator formula is $\sqrt{\text{terms}^2 / \text{SHOTS}} * \sqrt{2}$ (evidence/scripts/ablation_a_b.py) — 3 Pauli terms sharing 2000 shots/quadrature, per-shot variance ≤ 1 , **times an explicit sqrt(2) for combining both quadratures** (matching how the measured side is computed via `hypot(sem_re, sem_im)`). Omitting the `sqrt(2)` gives ≈ 0.067 , a $\sim 40\%$ miss — C4 audit caught this ambiguity; the formula above is the corrected, complete description.
- **R014 vs R008 damping** — 0.342 simulated vs 0.179 measured. Different circuits at different budgets; the comparison is qualitative and is labelled as such.
- **95% CIs** — CONVENTIONS §7 fixes these as $\pm 1.96 \cdot \sigma$. R019 tabulates both σ and CI; check they are consistent.

Known weak points — flag these before C4 does

1. **R017 is confounded** and marked NOT SLIDE-READY. The ideal-Trotter baseline it compares against is itself broken on the reduced grid (misses 2 of 4 lines before any noise). Recommendation: do not cite it; either re-run at Part-B fidelity or drop it.
2. **R008's <Q> line was withdrawn** (BUGLOG B04) — the number was measuring $3\langle Z_z \rangle$, not $\langle Q \rangle$. If any draft slide cites a hardware $\langle Q \rangle$, it is citing a retracted number.
3. **Headline budget vs side studies.** All reported Part A/B numbers come from ONE 256,000-shot record set (CONVENTIONS §5). The 12-seed study (3,072,000 shots), scaling study, noise study and ablation arms are *separately declared*. Any slide saying "we used N shots" must say which N it means.
4. **Hardware spans 13 jobs total** — 7 on our own account (R008; R018 x3; R023; R026; plus one accidental 2-shot submission logged in BUGLOG B05) and 6 run independently by a teammate on a different account, including `ibm_miami` which we cannot reach (R031). All of it supports robustness claims only, never a Part A/B headline — the Part A/B numbers are 100% simulator (R009-R022).
5. **Going-further: 6 of 18 items have committed results, 1 is excluded, 11 untouched.** DONE: variance budgeting (R029), chi/chi_Q correlation + correlated

bootstrap (R028, covering both its listings), rank selection (R027 — an honest NEGATIVE: the derived criterion does not beat the magic number, though it does no harm either), sector post-selection (R036), reduced hardware instance (R023, extended to 4-site in R026). PARTIAL: "push until it breaks" (R022 — the T_max axis only; not shots, not a 5th near-degenerate level). EXCLUDED: label-survival-under-noise (R017, confounded — do not cite). NOT ATTEMPTED: the remaining 11, including readout mitigation. Separately, work BEYOND the going-further lists: Tracks B/C/E (R034/R032/R033), the QPE ablation (R030), AQC-Tensor compression (R035), and the SKQD pipeline (R037). Any claim of "we explored the extensions" must be scoped to exactly this list.

Banned-language checklist (rubric.md)

- "quantum advantage" — must not appear
- uncoded "for free" — the shadow route costs a measured $1.07\times$ variance premium (R021) and ≤ 2 extra 1q gates per system qubit; say so wherever "no extra circuits" appears
- causal claims from single runs — R008/R018 arms are single runs each
- "hardware" — permitted, since the go/no-go passed with a completed job (R008)